

ML-Embed: Inclusive and Efficient Embeddings for a Multilingual World

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Abstract

The development of high-quality text embeddings is increasingly drifting toward an exclusionary future, defined by three critical barriers: prohibitive computational costs, a narrow linguistic focus that neglects most of the world’s languages, and a lack of transparency from closed-source or open-weight models that stifles research. To dismantle these barriers, we introduce ML-Embed, a suite of inclusive and efficient models built upon a new framework: 3-Dimensional Matryoshka Learning (3D-ML). Our framework addresses the computational challenge with comprehensive efficiency across the entire model lifecycle. Beyond the storage benefits of Matryoshka Representation Learning (MRL) and flexible inference-time depth provided by Matryoshka Layer Learning (MLL), we introduce Matryoshka Embedding Learning (MEL) for enhanced parameter efficiency. To address the linguistic challenge, we curate a massively multilingual dataset and train a suite of models ranging from 140M to 8B parameters. In a direct commitment to transparency, we release all models, data, and code. Extensive evaluation on 430 tasks demonstrates that our models set new records on 9 of 17 evaluated MTEB benchmarks, with particularly strong results in low-resource languages, providing a reproducible blueprint for building globally equitable and computationally efficient AI systems.

1. Introduction

Text embeddings are a foundational component of modern AI, translating the richness of human language into numerical representations that dictate the performance, fairness,

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and accessibility of the downstream systems they enable, from semantic search to Retrieval-Augmented Generation (RAG) (Gao et al., 2023).

However, the paradigm for developing state-of-the-art embedding models has shifted toward repurposing massive decoder-based language models. While powerful, this trend is creating a critical **computational barrier** characterized by prohibitive training costs and immense memory footprints. This computational barrier exacerbates a growing linguistic barrier: as models become more resource-intensive, they become increasingly inaccessible to the broader research community and undeployable in resource-constrained environments where many of the world’s low-resource languages are spoken. While techniques like Matryoshka Representation Learning (MRL, Kusupati et al., 2022) offer partial relief by optimizing storage, they leave the immense burdens of training and inference untouched.

To dismantle this computational barrier, we introduce 3-Dimensional Matryoshka Learning (3D-ML), a unified framework built upon Matryoshka Layer Learning (MLL, Li et al., 2024a) and Matryoshka Representation Learning (MRL, Kusupati et al., 2022) while integrating a novel Matryoshka Embedding Learning (MEL) technique that addresses the critical challenge of parameter-heavy embedding layers by learning two factorized, low-rank matrices that are themselves structured for nested training. This provides significant parameter savings for both training and inference and offers flexible deployment options that balance efficiency with compatibility.

To validate the practical utility of 3D-ML, we applied it to the notoriously resource-intensive challenge of creating massively multilingual models—a domain that faces two further critical gaps. The first is a **linguistic challenge**: despite comprehensive benchmarks like MTEB (Muennighoff et al., 2023; Enevoldsen et al., 2025), research attention remains disproportionately focused on a few high-resource languages. As illustrated in Table 1, submissions to benchmarks for languages like Polish, Persian, and Vietnamese are orders of magnitude fewer than for English. The second is a **transparency challenge**: progress is stymied by a lack of openness, as many top-performing models (Zhang et al., 2025b; Lee et al., 2025b) are released as closed-source APIs or as open-weight models with no training transparency,

hindering reproducible research. Addressing these interconnected issues, we introduce ML-Embed, a family of efficient and inclusive models built with our 3D-ML framework and a new, massively multilingual dataset.

This work makes the following contributions:

- We propose MEL, a novel efficient training technique. Integrating MEL with MLL and MRL, we present a 3-Dimensional Matryoshka Learning framework, providing end-to-end efficiency for the training, inference, and storage of embedding models.
- We demonstrate that efficiency and inclusivity can drive superior performance. ML-Embed-8B establishes new state-of-the-art results on 9 of 17 MTEB benchmarks. Crucially, we achieve massive gains in historically underserved languages—such as a +22.89 point improvement on Polish and +6.88 on Vietnamese—proving that equitable performance need not come at the cost of efficiency.
- In a direct counter to the trend of closed development, we release our comprehensive multilingual dataset, all model weights, and training code, providing a fully reproducible blueprint for building globally equitable AI systems¹.

2. Related Work

2.1. Efficient Representation Learning

Matryoshka Representation Learning (Kusupati et al., 2022) optimizes d -dimensional embeddings by applying loss functions at $O(\log(d))$ embedding sizes, facilitating adaptive application in downstream tasks with varying dimension requirements. Recent extensions such as ESE (Li et al., 2025b) improve MRL by applying principal component analysis to condense more essential information into the initial embedding dimensions and model layers, while Matryoshka-Adaptor (Yoon et al., 2024) and SMEC (Zhang et al., 2025a) employ additional MLP layers to reduce the embeddings to lower dimensions. Other methods, such as Flextron (Cai et al., 2024) and MatFormer (Devvrit et al., 2024), also enable flexible model sizes by pruning attention heads or MLP dimensions at inference time. However, these methods often introduce structural modifications (e.g., routing mechanisms) that may reduce compatibility and complicate deployment.

In terms of training, existing matryoshka optimization methods focus on representation flexibility but do not reduce

¹Code: <https://github.com/codefuse-ai/CodeFuse-Embeddings>. Model and data: <https://huggingface.co/collections/codefuse-ai/codefuse-embeddings>.

Table 1. Number of models with complete results on MTEB benchmarks. While Multilingual and English have become popular testbeds for embedding models, some languages - especially Polish, Japanese, Vietnamese, and Persian - receive far less attention.

Benchmark	Models
Multilingual	146
English	154
European	122
Indic	111
Scandinavian	41
Chinese	43
German	90
French	111
Japanese	11
Korean	95
Dutch	87
Polish	1
Russian	95
Persian	22
Vietnamese	17

trainable parameters or the overall training cost, making them ill-suited to data-constrained or compute-constrained training scenarios. In these settings, parameter-efficient fine-tuning methods such as LoRA (Hu et al., 2022) are used instead, which reduces the model’s trainable parameters by decomposing the *update* in weight matrices into low-rank ones. Numerous variants of LoRA have also been proposed, such as QLoRA (Dettmers et al., 2023) that combines LoRA and quantization, AdaLoRA (Zhang et al., 2023a) that adaptively allocates the parameter budget according to the importance of weight matrices, and RaSA (He et al., 2025) that partially shares LoRA parameters across model layers. Nevertheless, all these methods require the entire model to be loaded at inference time, limiting their utility for resource-constrained deployment where reduced memory footprints are required.

2.2. Multilingual Embedding Models and Benchmarks

The previous generation of encoder-based embedding models witnessed a proliferation of massively multilingual embedding models supporting hundreds of languages, represented by XLM-R (Conneau et al., 2020), mDeBERTaV3 (He et al., 2023), mBART (Liu et al., 2020), and mT5 (Xue et al., 2021). Recently, decoder-based embedding models have become the dominant paradigm, benefiting from their extensive capabilities acquired during large-scale pre-training, as verified by state-of-the-art models such as E5-Mistral (Wang et al., 2024a), NV-Embed (Lee et al., 2025a), Qwen3-Embedding (Zhang et al., 2025b), and Gemini-Embedding (Lee et al., 2025b).

However, this advancement has been accompanied by a shift toward English-centric evaluation. This is evidenced in MTEB (Muennighoff et al., 2023), which has been es-

tablished as one of the most recognized text embedding benchmarks, covering over 500 evaluation tasks and more than 250 languages (Enevoldsen et al., 2025). Yet, in reality, the MTEB leaderboards exhibit significant linguistic bias. For instance, in the MTEB-Multilingual benchmark, 35 out of the 131 tasks focus exclusively on English, potentially obscuring a model’s true multilingual efficacy. Furthermore, as illustrated in Table 1, many language-specific benchmarks receive disproportionately less attention compared with the English or Multilingual benchmarks².

This disparity is exacerbated by the fact that many top-performing multilingual embedding models - such as Qwen3-Embedding (Zhang et al., 2025b), Gemini-Embedding (Lee et al., 2025b), and EmbeddingGemma (Vera et al., 2025) - are either closed-source APIs or open-weight only without training transparency. KaLM-Embedding (Zhao et al., 2025) represents one of the few exceptions with transparency in training data, but focuses exclusively on the Multilingual leaderboard and is not evaluated on the aforementioned language-specific benchmarks that are critical for truly global applications.

3. Method: 3D Matryoshka Learning

Creating truly accessible and scalable embedding models requires tackling efficiency bottlenecks across the entire model lifecycle: from the high costs of **training**, to the computational demands of **inference**, and finally to the footprint of **storage**. To this end, we propose **3-Dimensional Matryoshka Learning (3D-ML)**, a unified framework that generalizes the principle of nested structures to provide comprehensive efficiency. 3D-ML simultaneously targets all three stages by optimizing along three corresponding axes: model parameters, computational depth, and representation size. This is achieved through a trio of integrated techniques: 1) **Matryoshka Embedding Learning (MEL)** reduces trainable and total parameters for efficient training and inference; 2) **Matryoshka Layer Learning (MLL)** enables flexible model depth for efficient inference; 3) **Matryoshka Representation Learning (MRL)** produces variable-size representation dimensions for efficient storage. Figure 1 provides a conceptual illustration of this framework.

3.1. Matryoshka Embedding Learning (MEL) for Parameter Efficiency

The embedding layer, which maps vocabulary tokens to dense vectors, often constitutes a disproportionately large share of a model’s parameters, especially in smaller models and multilingual models with a large vocabulary. For instance, in an embedding model trained from Qwen3-

0.6B (Yang et al., 2025), the embedding layer accounts for 1/4 of the total parameters. MEL addresses this by learning the embedding matrix in a factorized, low-rank form that is itself structured for nested training. Crucially, unlike low-rank update methods such as LoRA (Hu et al., 2022), MEL reduces not only *trainable parameters*, but also *total parameters* so that inference efficiency is also improved.

Let the base model’s original embedding matrix be $E \in \mathbb{R}^{v \times d_{\text{model}}}$, where v is the vocabulary size and d_{model} is the model’s hidden dimension. Prior to fine-tuning, we initialize two smaller matrices, E_A and E_B , using a truncated Singular Value Decomposition (SVD) of E . We compute $U, S, V^T = \text{SVD}(E)$ and select the top- r singular values and vectors to form $U_r \in \mathbb{R}^{v \times r}$, $S_r \in \mathbb{R}^{r \times r}$, and $V_r^T \in \mathbb{R}^{r \times d_{\text{model}}}$. The trainable matrices are then initialized as:

$$E_A \leftarrow U_r S_r \in \mathbb{R}^{v \times r} \quad \text{and} \quad E_B \leftarrow V_r^T \in \mathbb{R}^{r \times d_{\text{model}}}. \quad (1)$$

The full embedding matrix is approximated by their product, $E \approx E_A E_B$. During fine-tuning, only E_A and E_B are updated instead of a full $v \times d_{\text{model}}$ matrix, reducing trainable parameters and memory requirements.

To embed the Matryoshka principle, during each training forward pass, we dynamically sample a sub-rank $r' < r$ from a predefined set (e.g., $\{64, 128, 256, 512, 1024\}$). The forward pass then uses only the first r' components of the factorized matrices:

$$E_{\text{effective}} = E_A[:, :r'] E_B[:, :r']. \quad (2)$$

This forces the model to prioritize the most critical information within the initial dimensions of the factorized space.

At inference time, MEL offers two modes:

- **Compatibility Mode:** We compute the final trained embedding matrix $E_{\text{trained}} = E_A E_B$. This results in a standard embedding layer, requiring no changes to existing inference infrastructure while still benefiting from the regularized training via low-rank factorization.
- **Efficiency Mode:** For maximum resource savings, we can deploy the model with a highly compressed embedding layer. After training, we can either use the trained factorized matrices E_A and E_B directly (at rank r) or re-factorize the full matrix $E_{\text{trained}} = E_A E_B$ to an even smaller rank $r' \ll r$ for aggressive compression. Let the new factorized matrices be $E'_A \in \mathbb{R}^{v \times r'}$ and $E'_B \in \mathbb{R}^{r' \times d_{\text{model}}}$. This approach yields two key benefits. First, it drastically reduces the storage space: instead of storing a dense $v \times d_{\text{model}}$ matrix, we only store $v \times r' + r' \times d_{\text{model}}$ parameters. For a large vocabulary V and a small rank r' , this represents a

²All references to MTEB leaderboards in this manuscript refer to the snapshot acquired on January 22nd, 2026.

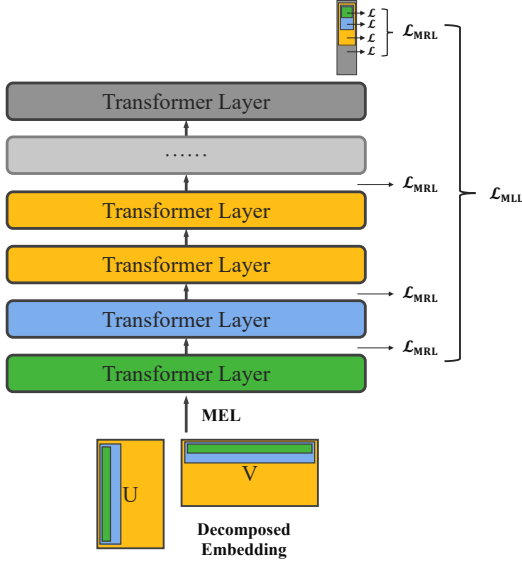


Figure 1. The 3D-ML framework provides comprehensive efficiency by applying nested learning principles across model parameters (MEL), depth (MLL), and representation dimensions (MRL).

substantial reduction. Second, it can improve computational efficiency. A standard embedding lookup for a sequence of tokens involves gathering rows from the large $v \times d_{\text{model}}$ matrix. With factorization, this becomes a two-step process: a fast lookup in the “tall-and-skinny” matrix E'_A followed by a matrix multiplication with the “short-and-wide” matrix E'_B . This is particularly advantageous for on-device deployment where memory is the primary constraint.

3.2. Matryoshka Layer Learning (MLL) for Inference Efficiency

The computational cost of Transformer-based models scales with model depth. MLL is designed to produce models that can be dynamically and efficiently truncated to shallower depths without significant performance degradation.

Instead of applying the training loss only at the final layer’s output, MLL applies it at multiple, pre-defined intermediate layers. Let $\mathcal{L}_{\text{layers}} = \{l_1, l_2, \dots, l_k, L\}$ be a set of selected layer indices, where L is the index of the final layer. For our experiments, we use a logarithmically spaced set of layers (e.g., $\{1, 2, 4, 8, 16, 32\}$) plus the model’s final layer. For each layer $l \in \mathcal{L}_{\text{layers}}$, we extract its hidden state output, h_l . To maintain representational consistency across depths, we pass each h_l through the model’s final layer normalization, LN_{final} , before using it to compute the loss.

This “early-exit” style training ensures that shallower versions of the model are also effective embedders. At inference time, this provides unparalleled flexibility: one can

deploy a smaller, faster model by simply taking the first l layers of the full model, where $l \in \mathcal{L}_{\text{layers}}$. This avoids the need for re-training or complex pruning, enabling seamless adaptation to varying computational budgets.

3.3. Unifying the Framework with Matryoshka Representation Learning (MRL)

The final dimension of our framework is Matryoshka Representation Learning (MRL; Kusupati et al., 2022), which optimizes embeddings for variable-dimension storage. MRL trains a model such that prefixes of the final embedding vector are themselves effective, lower-dimensional representations.

In 3D-ML, MRL is not a separate step but is deeply integrated with MLL. For each selected MLL layer $l \in \mathcal{L}_{\text{layers}}$, we apply a contrastive loss not just on the full-dimensional output representation, but on a nested set of its prefixes. Let $\mathcal{D}_{\text{mrl}}$ be the set of MRL dimensions (e.g., $\{8, 16, 32, \dots, d_{\text{model}}\}$). Let $\text{proj}_d(v)$ denote the projection of a vector v to its first d dimensions. The total loss for a given layer l is a sum of losses over these dimensions.

The unified 3D-ML objective function combines all three components. The total loss is summed over all selected MLL layers and all MRL dimensions. Let $h_l(q)$ and $h_l(d)$ be the hidden states from layer l for a query q and document d , respectively. The final representation for a given MRL dimension d' is $v_{l,d'}(\cdot) = \text{proj}_{d'}(\text{LN}_{\text{final}}(h_l(\cdot)))$. The overall objective is:

$$\mathcal{L}_{\text{3D-ML}} = \sum_{l \in \mathcal{L}_{\text{layers}}} \sum_{d' \in \mathcal{D}_{\text{mrl}}} c_{l,d'} \mathcal{L}_{\text{cl}}(q_i, d_i^+, \{d_{i,j}^-\}_{j=1}^n; v_{l,d'}), \quad (3)$$

where $c_{l,d'}$ is the loss weight coefficient for layer l and dimension d' , and the contrastive learning loss $\mathcal{L}_{\text{cl}}$ for a given representation function $v_{l,d'}$ is defined as:

$$-\log \frac{e^{s(v_{l,d'}(q_i), v_{l,d'}(d_i^+))/\tau}}{e^{s(v_{l,d'}(q_i), v_{l,d'}(d_i^+))/\tau} + \sum_{j=1}^n e^{s(v_{l,d'}(q_i), v_{l,d'}(d_{i,j}^-))/\tau}}. \quad (4)$$

Here, $s(\cdot, \cdot)$ is cosine similarity, τ is a temperature hyperparameter, d_i^+ is a positive document for query q_i , and $\{d_{i,j}^-\}$ are hard negative documents for query q_i . This unified loss ensures that the model learns representations that are simultaneously efficient in terms of parameters (via MEL), depth (via MLL), and storage (via MRL).

3.4. Practical Deployment and Compatibility

A core design principle of the 3D-ML framework is its focus on practical deployment and compatibility with existing ecosystems, ensuring that its efficiency gains are not merely theoretical but easily accessible to practitioners. Each com-

ponent is designed for minimal friction:

- **MRL (Storage):** The benefits of Matryoshka Representation Learning are straightforward to leverage. Truncating the final embedding vectors is a simple post-processing step that is natively supported in popular libraries like SENTENCE-TRANSFORMERS (Reimers & Gurevych, 2019) via a single parameter, requiring no changes to the codebase.
- **MLL (Inference):** Matryoshka Layer Learning is similarly user-friendly and fully compatible with the Hugging Face ecosystem. Deploying a faster, shallower model is as simple as modifying the `num_hidden_layers` parameter in the model’s configuration file, which drives the AutoModel class from TRANSFORMERS (Wolf et al., 2020) to load only the first n layers and ignore the remaining weights.
- **MEL (Parameters):** As described in Section 3.1, Matryoshka Embedding Learning offers a flexible trade-off between convenience and maximum efficiency. In **compatibility mode**, the factorized matrices (E_A , E_B) are multiplied into a standard embedding matrix before release, making the model indistinguishable from a standard Transformer decoder, ensuring seamless integration into any inference pipeline without code changes. For users prioritizing a minimal memory footprint, the **efficiency mode** allows for deploying the low-rank factorized matrices directly, enabling significant parameter reduction with only minor adjustments to the modeling file.

This comprehensive focus on deployability makes 3D-ML a practical blueprint for building and sharing highly efficient models, lowering the barrier to entry for a wide range of users, from large-scale production systems to resource-constrained research environments.

4. Training Data

A cornerstone of our work is the compilation of a vast and diverse training corpus designed to foster both linguistic inclusivity and broad task competency. We aggregate data from 121 publicly available sources, creating a collection of 50 million training samples that span 282 natural languages (as identified by ISO-639-3 codes) and over 40 programming languages. Crucially, our data curation process is driven by real-world data availability rather than optimizing for specific benchmarks. For instance, our dataset contains substantial data for Spanish and Arabic, which are the 3rd and 7th most represented languages in our corpus (Figure 3), despite these languages lacking dedicated benchmarks in MTEB (see Table 1). This approach, which also includes a long tail of low-resource languages and a significant volume

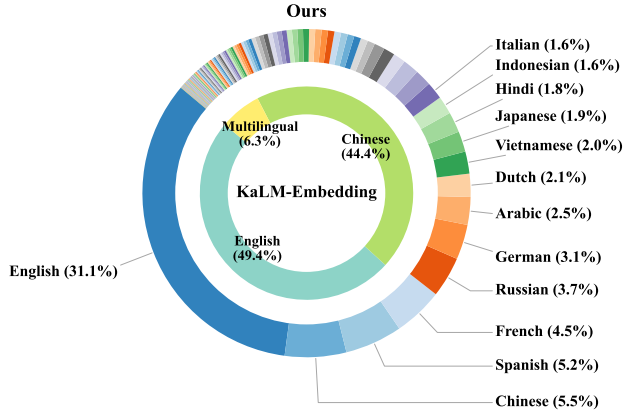


Figure 2. Comparison between the language distribution of our training data (outer circle) and KaLM-Embedding (inner circle). KaLM-Embedding’s data is only annotated with three labels, while ours are annotated with specific languages.

of code, aims to build a model with truly global utility and directly contrasts recent open-source datasets such as that released by KaLM-Embedding, which is heavily skewed towards English and Chinese (Figure 2). We provide a more comprehensive linguistic breakdown of our dataset in Appendix A.

The functional diversity of our dataset is equally critical for training a general-purpose embedding model. As shown in Figure 7 in Appendix A, our collection encompasses a wide spectrum of tasks, ranging from retrieval-focused question answering and bitext mining to classification-oriented sentiment analysis and intent/domain classification.

To leverage this heterogeneity within a unified contrastive learning framework, we follow prior work (Lee et al., 2025a; Zhang et al., 2025c) and consolidate all data into three canonical formats: *retrieval*, *clustering*, and *two-way classification*. This consolidation allows the model to learn a versatile embedding space by optimizing a single, consistent objective across disparate data sources and task structures. For the retrieval format, data consists of (query, positive document, hard negatives) tuples. We leverage both in-batch negatives, where other documents in a mini-batch serve as negatives, and explicitly provided hard negatives (mined using Qwen3-Embedding-8B) to create a challenging and efficient training signal. For the clustering format, which also ingests multi-class classification tasks, tuples are formed by sampling an anchor, a positive example from the same class, and a hard negative from a different class. Finally, the two-way classification format directly uses class labels, where a given text serves as the anchor, the corresponding label text is the positive, and the opposite label text is the negative. For both clustering and classification, only hard negatives are utilized to avoid introducing false negatives from in-batch samples.

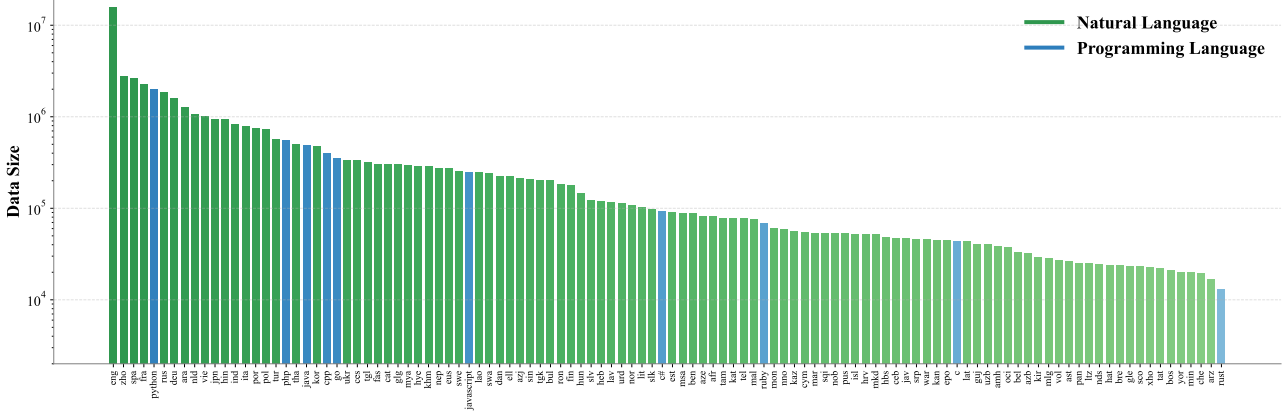


Figure 3. Top-100 natural languages and top-10 programming languages in our training data.

To maximize the utility of this diverse corpus, we adopt a two-stage training strategy following previous works (Lee et al., 2025a; Zhang et al., 2025b). The first stage focuses on building a robust semantic foundation by training on a large-scale subset of retrieval datasets, totaling 27 million samples. This phase imbues the model with a strong general understanding of semantic similarity. In the second stage, we conduct fine-tuning on a sampled mixture of 8.3 million samples from all data sources, applying task-specific instructions to the queries. This stage sharpens the model’s ability to handle the nuances of diverse downstream applications like classification, reranking, and paraphrase detection.

5. Experiments

5.1. Experimental Setting

Model We present a series of 6 models, all trained on identical data in exactly the same order: 140M, 330M, 600M, 1.7B, 4B, 8B. All models are fine-tuned from Qwen3 causal LLMs (Yang et al., 2025), where 600M, 1.7B, 4B, and 8B models correspond to models of the same size in the Qwen3 family, while 140M and 330M models are pruned from the 600M model after training. Following existing embedding models based on the Qwen3 family (Zhang et al., 2025b;c), we maintain the causal attention in the models and use EOS token representation as the sequence embedding.

Training Inspired by NV-Embed (Lee et al., 2025a) and Qwen3-Embedding (Zhang et al., 2025b), we train the models in two stages. In the first stage, we use only some of the largest retrieval datasets, and do not apply instructions to the data, aiming to inject the causal models with basic capabilities of converting texts into semantic embeddings that are usable by downstream tasks. In the second stage, we sample from all data sources, and apply instructions to queries, enabling a more nuanced understanding of different semantic representation tasks such as retrieval, classification,

and paraphrase detection. More details about the amount of training data and hyperparameter settings are given in Appendix C, where we also demonstrate the empirical benefits of the two-stage training design.

Evaluation We evaluate the models on 17 MTEB benchmarks: Multilingual (Enevoldsen et al., 2025), English (Enevoldsen et al., 2025), Code (Enevoldsen et al., 2025), Medical, European (Enevoldsen et al., 2025), Scandinavian (Enevoldsen et al., 2024), Indic (Enevoldsen et al., 2025), German (Wehrli et al., 2023), French (Ciancone et al., 2024), Korean, Polish (Poswiata et al., 2024), Chinese (Xiao et al., 2023), Japanese (Li et al., 2026), Dutch (Banar et al., 2025), Russian (Snegirev et al., 2025), Persian (Zinvandi et al., 2025), and Vietnamese (Pham et al., 2026), totaling 430 tasks across ten types: retrieval, reranking, classification, clustering, pair classification, multilabel classification, STS, instruction reranking, bitext mining, and summarization. More details on these benchmarks and tasks are given in Appendix B. For comparison, we report the previous top-1 score and top-5 score on each benchmark’s leaderboard. We also compare with individual models, specifically those from the Qwen3-Embedding (Zhang et al., 2025b) and EmbeddingGemma (Vera et al., 2025) families.

5.2. MTEB Results

We present the main results in Table 2, comparing the ML-Embed family against the top-performing models on 17 MTEB benchmarks. Our largest model, ML-Embed-8B, establishes new state-of-the-art (SOTA) scores on a remarkable 9 out of 17 benchmarks, demonstrating the effectiveness of our multilingual training corpus.

Critically, these SOTA results are concentrated in benchmarks for languages historically underserved by the research community, directly addressing the linguistic challenge outlined in Section 1. For instance, on the Polish benchmark,

Table 2. Comparison of our models against previous top-1 and top-5 performance on 17 MTEB benchmark leaderboards. The number of tasks in each benchmark is given in ^(superscript). The specific metrics are consistent with the main metrics used by MTEB (e.g., NDCG@10 for retrieval tasks and accuracy for classification tasks).

Model	Multi. ⁽¹³¹⁾	English ⁽⁴¹⁾	Code ⁽¹²⁾	Medical ⁽¹²⁾	European ⁽⁷³⁾	Scan. ⁽²⁸⁾	Indic ⁽²⁰⁾	German ⁽¹⁹⁾	French ⁽²⁵⁾
<i>Top results on the leaderboard</i>									
<i>Top-1</i>	72.32	75.97	80.75	66.55	63.60	65.56	70.15	59.96	70.37
<i>Top-5</i>	69.45	74.61	76.00	63.83	62.32	62.01	67.39	55.72	67.25
<i>Ours</i>									
<i>8B</i>	66.79	73.26	80.28	62.91	68.00	69.49	76.76	66.43	71.91
<i>4B</i>	65.80	72.89	79.95	62.05	67.53	67.88	75.15	65.49	70.97
<i>1.7B</i>	63.70	71.19	78.90	59.88	65.47	65.79	72.58	63.99	68.94
<i>0.6B</i>	61.30	70.01	77.27	57.58	63.40	62.00	66.11	61.58	66.64
<i>330M</i>	58.06	67.99	73.94	55.91	59.80	59.22	61.87	58.48	63.59
<i>140M</i>	49.81	60.35	61.78	48.28	50.40	47.95	46.37	46.66	52.83
<i>Results continued for remaining languages and average</i>									
Model	Korean ⁽⁶⁾	Polish ⁽¹⁷⁾	Chinese ⁽³²⁾	Japan. ⁽²⁸⁾	Dutch ⁽⁴⁰⁾	Russian ⁽²³⁾	Persian ⁽⁵²⁾	Viet. ⁽⁵⁰⁾	Average
<i>Top results on the leaderboard</i>									
<i>Top-1</i>	77.01	50.95	78.52	73.18	58.38	74.16	71.58	54.74	68.46
<i>Top-5</i>	69.06	n.a.	74.87	69.51	56.23	69.39	65.26	52.37	65.95
<i>Ours</i>									
<i>8B</i>	74.84	73.84	67.22	77.81	62.64	69.24	71.12	61.62	70.24
<i>4B</i>	73.32	73.14	66.55	76.65	61.42	67.93	69.94	61.20	69.29
<i>1.7B</i>	72.33	71.12	64.87	74.53	59.83	67.08	68.35	60.27	67.58
<i>0.6B</i>	68.74	68.13	62.64	71.30	56.59	63.35	65.32	57.85	64.69
<i>330M</i>	61.71	65.65	59.38	66.00	54.04	60.12	60.16	54.17	61.18
<i>140M</i>	53.07	51.22	51.02	55.97	43.89	47.24	52.50	43.67	50.76

our model achieves a score of 73.84, a staggering +22.89 point improvement over the previous best model. Similarly, we set new records on Vietnamese (+6.88), Indic (+6.61), German (+6.47), Japanese (+4.63), Dutch (+4.26), French (+1.54), and the aggregated benchmarks of Scandinavian (+3.93) and European (+4.40). This demonstrates that our data curation and training methodology successfully produce models with globally equitable performance.

On the highly competitive English and Multilingual benchmarks, our models also perform comparably to the top-5 models on the leaderboard, validating our approach as a strong foundation for general-purpose embeddings. Furthermore, the results exhibit a clear and consistent scaling trend: performance reliably improves with model size across all benchmarks. This indicates that our training recipe is robust and provides a scalable blueprint for developing even more powerful models in the future.

5.3. Comparison with Similar-Sized Models

In Table 3, we compare our 0.3B and 0.6B models with EmbeddingGemma-0.3B and Qwen3-Embedding-0.6B, respectively. For these two models, we use their public results in the MTEB repository when available, and evaluate the remaining tasks using the same prompts as those used for evaluating our models. The results are similar to those on the MTEB leaderboard: our models demonstrate superior performance on the less-attended Code, Scandinavian, German,

French, Korean, Polish, Japanese, Dutch, and Vietnamese benchmarks, while underperforming on English, Chinese, and Multilingual benchmarks.

5.4. Ablation Studies

To dissect the individual contributions of our proposed efficiency methods, we conduct a series of ablation studies on the English subset using the 0.6B model.

MLL and MEL Synergy First, we investigate the interplay between Matryoshka Layer Learning (MLL) and Matryoshka Embedding Learning (MEL). As shown in Figure 4, we compare four settings: (1) a single baseline, evaluated at different exit layers; (2) baselines of individually trained models at different depths; (3) a single model trained with MLL and evaluated at different exit layers; and (4) a fourth model combining MLL and MEL. MLL alone presents a classic trade-off: it enables training a single, depth-flexible model for the computational cost of one, but the resulting shallower models slightly underperform individually trained counterparts. However, the introduction of MEL dramatically alters this dynamic. By significantly reducing the parameter count of the embedding layer, MEL allows for a much deeper model at the same parameter budget. For example, our MLL+MEL model with 4 layers has the same parameter count (170M) as a 1-layer baseline model but achieves a 15-point higher score. At equivalent performance levels, the MLL+MEL model is 3x smaller, confirming the

Table 3. Comparison of our models with EmbeddingGemma and Qwen3-Embedding. The number of tasks in each benchmark is given in (superscript).

Model	Multi. ⁽¹³¹⁾	English ⁽⁴¹⁾	Code ⁽¹²⁾	Medical ⁽¹²⁾	European ⁽⁷³⁾	Scan. ⁽²⁸⁾	Indic ⁽²⁰⁾	German ⁽¹⁹⁾	French ⁽²⁵⁾
<i>0.3B</i>									
<i>EmbedGemma</i>	<u>61.15</u>	<u>69.67</u>	68.76	51.24	<u>62.50</u>	54.39	<u>66.11</u>	56.28	61.90
<i>Ours</i>	58.06	67.99	<u>73.94</u>	<u>55.91</u>	59.80	<u>59.22</u>	61.87	<u>58.48</u>	<u>63.59</u>
<i>0.6B</i>									
<i>Qwen3-Embed</i>	<u>64.34</u>	<u>70.47</u>	75.42	<u>60.16</u>	<u>63.91</u>	60.99	<u>66.53</u>	59.45	63.01
<i>Ours</i>	61.30	70.01	<u>77.27</u>	57.58	<u>63.40</u>	<u>62.00</u>	66.11	<u>61.58</u>	<u>66.64</u>
<i>Results continued for remaining languages and average</i>									
Model	Korean ⁽⁶⁾	Polish ⁽¹⁷⁾	Chinese ⁽³²⁾	Japan. ⁽²⁸⁾	Dutch ⁽⁴⁰⁾	Russian ⁽²³⁾	Persian ⁽⁵²⁾	Viet. ⁽⁵⁰⁾	Avg.
<i>0.3B</i>									
<i>EmbedGemma</i>	58.24	64.70	50.40	60.82	50.98	<u>64.57</u>	<u>67.11</u>	43.45	59.55
<i>Ours</i>	<u>61.71</u>	<u>65.65</u>	<u>59.38</u>	<u>66.00</u>	<u>54.04</u>	60.12	<u>60.16</u>	<u>54.17</u>	<u>61.18</u>
<i>0.6B</i>									
<i>Qwen3-Embed</i>	65.29	67.42	<u>66.71</u>	67.28	54.27	<u>64.20</u>	62.88	56.01	64.02
<i>Ours</i>	<u>68.74</u>	<u>68.13</u>	62.64	<u>71.30</u>	<u>56.59</u>	63.35	<u>65.32</u>	<u>57.85</u>	<u>64.69</u>

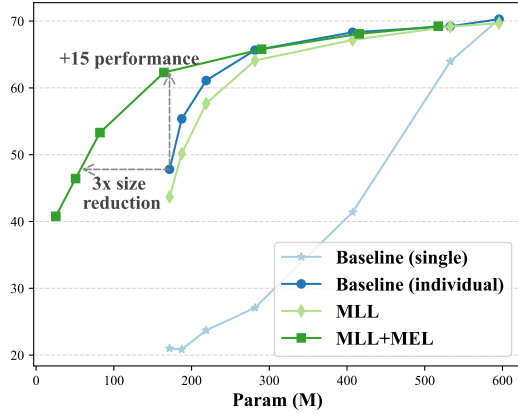


Figure 4. Ablation results of models pruned to different depths on MTEB-English. Each point on the baseline (individual) curve represents an individual trained model, while points on the Baseline (single), MLL, and MLL+MEL curves are models of different depths pruned from a single trained model.

powerful synergy between these two techniques for creating parameter-efficient models.

Robustness of MEL Next, we isolate the effect of MEL on inference-time compression. We compare a baseline model against two variants: one trained with a factorized embedding layer at rank 512 (“Decomposition-only”) and another additionally trained with the nested rank objective of MEL. At inference, we apply SVD to each model’s embedding matrix and evaluate performance at progressively smaller ranks. The results in Figure 5 are stark. The baseline model is extremely brittle - its performance collapses catastrophically (from 69.68 to 53.25) with even minor rank reduction (from 1024 to 960). The decomposition-only model is more robust, as the low-rank structure acts as a training regularizer even though the model is not trained

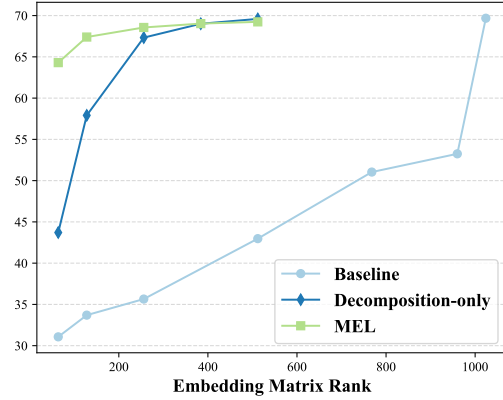


Figure 5. Ablation results of decomposing the embedding layer to varying ranks at inference time. Baseline model is trained with the original embedding. Decomposition-only model is trained with decomposed embedding (at rank 512). MEL model is trained with decomposed embedding plus Matryoshka Embedding Learning.

to concentrate information on the leading ranks. Notably, it achieves almost identical performance to the baseline (69.60) at rank 512, demonstrating the redundancy in the embedding matrix. The MEL-trained model demonstrates superior robustness against decomposition, declining much more slowly as the rank diminishes, retaining a strong score of 64.30 even when compressed to a rank of just 64. This confirms that MEL is highly effective at producing models that are robust to aggressive, post-hoc compression.

Data Comparison To isolate the impact of our curated training corpus, we conduct a head-to-head comparison against the recently released KaLM-Embedding finetuning data (Zhao et al., 2025). Starting from the same stage-1 checkpoint, we finetune two identical 0.6B models using our stage-2 data and the similar-sized KaLM-Embedding data, respectively. The results, shown in Figure 6, demon-

Table 4. Comparison of baseline training, pruned model training, and 3D-ML on EuroBERT backbone.

Model	Multi.	English	Code	Medical	European	Scan.	Indic	German	French
<i>Baseline (210M)</i>	59.86	68.52	64.03	55.93	59.83	54.43	55.36	62.90	62.90
<i>Pruned (120M)</i>	46.08	54.65	35.44	39.52	44.79	40.75	42.90	47.69	47.94
<i>3D-ML (120M)</i>	56.94	66.13	59.76	51.83	56.75	51.36	50.53	59.84	58.96
<i>Results continued for remaining languages and average</i>									
Model	Korean	Polish	Chinese	Japan.	Dutch	Russian	Persian	Viet.	Avg.
<i>Baseline (210M)</i>	61.01	64.63	59.37	71.82	52.26	58.44	60.00	55.22	60.38
<i>Pruned (120M)</i>	42.44	47.73	43.48	52.86	35.39	42.29	48.30	37.48	44.10
<i>3D-ML (120M)</i>	58.27	59.13	56.18	67.56	47.84	54.50	58.24	51.24	56.77

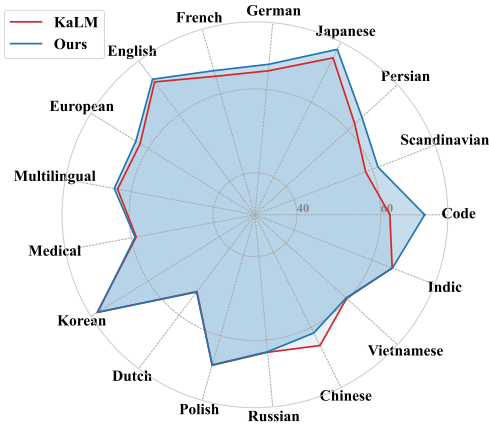


Figure 6. Comparison between 0.6B models trained on our data and KaLM-Embedding data.

strate the distinct advantages of our data curation strategy. Our model achieves superior performance on 9 out of 17 benchmarks, including the composite Multilingual, European, and Scandinavian sets, as well as English, French, German, Japanese, and Persian. The most significant lead is on the Code benchmark, highlighting our data’s wider domain coverage. While the KaLM-Embedding data produces a stronger model for Chinese - an expected outcome given its heavy concentration on Chinese and English data (Figure 2) - our dataset achieves on-par performance across seven other benchmarks, including Korean, Polish, Dutch, Indic, Russian, and Vietnamese. This outcome confirms that our focus on linguistic diversity yields a more globally robust model, trading hyper-specialization in a single language for broader competence.

Generalization to Different Backbones To verify the effectiveness of 3D-ML, we conduct additional experiments using EuroBERT-210M (Boizard et al., 2025) as the backbone. We train three models on the stage-2 data described in Section 4: 1) baseline finetuning, 2) structural pruning

to 120M parameters followed by finetuning, and 3) 3D-ML training followed by structural pruning to 120M parameters. The results in Table 4 show that training with 3D-ML leads to a minimal performance drop compared with naive structural pruning, demonstrating the generalizability of 3D-ML.

In Appendix D, we provide additional experiments including 1) applying 3D-ML to training on specific languages, 2) applying MEL to language modeling, and 3) in-depth analysis of efficiency gains.

6. Conclusion

This paper addresses the critical challenges of computational cost, linguistic bias, and lack of transparency hindering the development of text embeddings. We introduce ML-Embed, a family of models built using our 3-Dimensional Matryoshka Learning (3D-ML) framework. 3D-ML integrates Matryoshka Embedding (MEL), Layer (MLL), and Representation (MRL) Learning to achieve comprehensive efficiency across the model lifecycle. Paired with a newly curated, massively multilingual open-source dataset, our approach demonstrates that efficiency and inclusivity can yield state-of-the-art performance. Our 8B model sets new records on 9 of 17 MTEB benchmarks, with dramatic improvements in historically understudied languages such as Polish (+22.89) and Vietnamese (+6.88). Ablation studies confirm the synergistic benefits of 3D-ML components, creating powerful models adaptable to diverse computational budgets. The studies also highlight the superiority of our curated data.

By releasing our models, data, and code, we offer a reproducible blueprint for building globally equitable and efficient AI systems, dismantling the transparency barrier. Our work paves the way for future research in scaling these techniques to larger models, expanding linguistic coverage, and deploying powerful embeddings on resource-constrained devices. We hope this work steers the field toward a more inclusive and accessible future for text representation learning.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Details on Training Data

Table 5. Natural language distribution in our training data (part1).

ISO Code	Language	Samples	ISO Code	Language	Samples	ISO Code	Language	Samples
eng	English	15,683,866	slk	Slovak	97,974	tat	Tatar	22,327
zho	Chinese	2,791,623	est	Estonian	91,102	bos	Bosnian	21,175
spa	Spanish	2,607,192	msa	Malay	87,873	yor	Yoruba	20,139
fra	French	2,251,567	ben	Bengali	87,787	min	Minangkabau	19,868
rus	Russian	1,848,954	aze	Azerbaijani	82,209	che	Chechen	19,518
deu	German	1,593,329	afr	Afrikaans	81,233	arz	Egyptian Arabic	16,783
ara	Arabic	1,264,371	tam	Tamil	78,384	lmo	Lombard	16,575
nld	Dutch	1,051,914	kat	Georgian	77,567	arg	Aragonese	16,500
vie	Vietnamese	1,005,512	tel	Telugu	77,362	bak	Bashkir	16,451
jpn	Japanese	935,603	mal	Malayalam	76,518	som	Somali	16,369
hin	Hindi	931,761	mon	Mongolian	59,851	als	Tosk Albanian	15,655
ind	Indonesian	817,131	nno	Norwegian Nynorsk	58,298	ido	Ido	15,613
ita	Italian	790,507	kaz	Kazakh	55,317	szl	Silesian	14,845
por	Portuguese	752,642	cym	Welsh	53,951	wuu	Wu Chinese	14,762
pol	Polish	730,068	mar	Marathi	53,803	new	Nepal Bhasa	14,714
tur	Turkish	565,158	sqi	Albanian	53,602	chv	Chuvash	13,759
tha	Thai	503,612	nob	Norwegian Bokmål	52,905	pnb	Western Panjabi	13,657
kor	Korean	478,665	pus	Pushto	52,753	fry	Western Frisian	13,649
ukr	Ukrainian	337,739	isl	Icelandic	52,546	snd	Sindhi	13,210
ces	Czech	330,256	hrv	Croatian	52,505	ori	Oriya	12,791
tgl	Tagalog	313,985	mkd	Macedonian	52,504	plt	Plateau Malagasy	12,717
fas	Persian	302,869	hbs	Serbo-Croatian	48,523	scn	Sicilian	12,694
cat	Catalan	301,984	ceb	Cebuano	47,408	kur	Kurdish	11,872
glg	Galician	301,386	jav	Javanese	47,283	sun	Sundanese	11,793
mya	Burmese	294,167	srp	Serbian	46,108	bar	Bavarian	11,193
hye	Armenian	288,622	war	Waray	45,348	yid	Yiddish	10,785
khm	Khmer	287,530	kan	Kannada	44,534	ckb	Central Kurdish	9,829
nep	Nepali	276,057	epo	Esperanto	44,266	fao	Faroese	9,825
eus	Basque	270,551	lat	Latin	43,335	ina	Interlingua	9,782
swe	Swedish	256,469	guj	Gujarati	40,184	gla	Scottish Gaelic	9,769
lao	Lao	244,750	uzb	Uzbek	39,951	bug	Buginese	9,662
swa	Swahili	241,497	amh	Amharic	38,763	que	Quechua	9,406
dan	Danish	224,504	oci	Occitan	37,413	bpy	Bishnupriya	9,400
ell	Modern Greek	223,254	bel	Belarusian	33,330	san	Sanskrit	8,730
azj	North Azerbaijani	213,046	azb	South Azerbaijani	31,815	lim	Limburgan	8,573
sin	Sinhala	207,903	kir	Kirghiz	29,319	hau	Hausa	8,435
tgk	Tajik	200,631	mlg	Malagasy	28,661	mai	Maithili	8,180
bul	Bulgarian	200,127	vol	Volapük	27,187	zsm	Standard Malay	8,179
ron	Romanian	184,034	ast	Asturian	26,004	ibo	Igbo	8,132
fin	Finnish	178,810	pan	Panjabi	25,096	vec	Venetian	8,121
hun	Hungarian	144,561	ltz	Luxembourgish	25,092	ilo	Iloko	7,968
slv	Slovenian	122,065	nds	Low German	24,713	asm	Assamese	7,042
heb	Hebrew	119,685	hat	Haitian	23,940	sah	Yakut	7,011
lav	Latvian	115,288	bre	Breton	23,931	arb	Standard Arabic	6,945
urd	Urdu	113,258	gle	Irish	23,148	sna	Shona	6,933
nor	Norwegian	107,023	sco	Scots	23,032	mlt	Maltese	6,911
lit	Lithuanian	101,937	xho	Xhosa	22,799	zul	Zulu	6,669

Table 6. Natural language distribution in our training data (part2).

ISO Code	Language	Samples	ISO Code	Language	Samples	ISO Code	Language	Samples
mzn	Mazanderani	6,352	tsn	Tswana	1,539	lvs	Standard Latvian	800
uig	Uighur	6,190	mwl	Mirandese	1,491	mag	Magahi	800
oss	Iron Ossetic	5,893	div	Dhivehi	1,387	mni	Manipuri	800
tuk	Turkmen	5,854	kbp	Kabiyè	1,349	mos	Mossi	800
ary	Moroccan Arabic	5,703	chm	Mari	1,238	nqo	N'Ko	800
wln	Walloon	5,408	ewe	Ewe	1,220	nus	Nuer	800
cdo	Min Dong Chinese	5,175	smo	Samoan	1,175	ory	Odia	800
npi	Nepali	5,156	tso	Tsonga	1,174	prs	Dari	800
nap	Neapolitan	4,778	fij	Fijian	1,122	quy	Ayacucho Quechua	800
ace	Achinese	4,758	bam	Bambara	1,061	sat	Santali	800
mrj	Western Mari	4,728	lin	Lingala	1,046	tpi	Tok Pisin	800
xmf	Mingrelian	4,712	nav	Navajo	1,028	tum	Tumbuka	800
pes	Iranian Persian	4,414	roh	Romansh	999	tzm	Central Atlas Tamazight	800
diq	Dimli	4,140	ssw	Swati	982	umb	Umbundu	800
apc	Levantine Arabic	4,079	awa	Awadhi	954	uzn	Northern Uzbek	800
wol	Wolof	4,068	pag	Pangasinan	954	ydd	Eastern Yiddish	800
pbt	Southern Pashto	3,796	cor	Cornish	938	yue	Yue Chinese	800
nso	Pedi	3,676	dzo	Dzongkha	928	dyu	Dyula	799
srd	Sardinian	3,614	udm	Udmurt	890	lua	Luba-Lulua	799
ban	Balinese	3,581	fon	Fon	883	twi	Twi	798
lij	Ligurian	3,487	kon	Kongo	873	aeb	Tunisian Arabic	793
hsb	Upper Sorbian	3,440	glv	Manx	864	kik	Kikuyu	761
acq	Ta'izzi-Adeni Arabic	3,188	tir	Tigrinya	851	tyv	Tuvinian	626
crh	Crimean Tatar	2,985	pms	Piemontese	842	ava	Avaric	588
mri	Maori	2,922	myv	Erzya	840	aym	Aymara	587
egl	Emilian	2,859	sag	Sango	827	krc	Karachay-Balkar	587
ars	Najdi Arabic	2,712	run	Rundi	825	ful	Fulah	581
grn	Guarani	2,705	aem	Mesopotamian Arabic	800	orm	Oromo	548
nya	Chichewa	2,607	aka	Akan	800	stq	Saterfriesisch	461
hif	Fiji Hindi	2,566	ayr	Central Aymara	800	lah	Lahnda	450
kas	Kashmiri	2,363	bem	Bemba	800	ton	Tonga	391
fur	Friulian	2,284	bho	Bhojpuri	800	mdf	Moksha	314
swh	Swahili	2,253	cjk	Chokwe	800	haw	Hawaiian	299
fil	Filipino	2,156	dik	Southwestern Dinka	800	nia	Nias	297
sme	Northern Sami	2,156	fuv	Nigerian Fulfulde	800	bis	Bislama	272
shn	Shan	2,098	gaz	West Central Oromo	800	alt	Southern Altai	250
sot	Southern Sotho	2,089	hne	Chhattisgarhi	800	srn	Sranan Tongo	204
kin	Kinyarwanda	2,031	kab	Kabyle	800	ven	Venda	194
lug	Ganda	1,994	kac	Kachin	800	kbd	Kabardian	172
pap	Papiamentu	1,974	kam	Kamba	800	xal	Kalmyk	122
cos	Corsican	1,949	kea	Kabuverdianu	800	din	Dinka	104
mhr	Eastern Mari	1,633	khk	Halh Mongolian	800	jam	Jamaican Creole English	100
bjn	Banjar	1,600	kmb	Kimbundu	800	kal	Kalaallisut	92
knc	Central Kanuri	1,600	kmr	Northern Kurdish	800	iku	Inuktitut	84
taq	Tamasheq	1,600	ltg	Latgalian	800	guc	Wayuu	52
kom	Komi	1,583	luo	Luo	800	chr	Cherokee	51
bod	Tibetan	1,563	lus	Lushai	800	ady	Adyghe	33

Table 7. Programming language distribution in our training data.

Language	Samples	Language	Samples
python	1,972,390	css	1,003
php	553,651	typescript	888
java	483,469	r	636
cpp	393,514	lisp	467
go	351,586	jsx	436
javascript	245,632	objective-c	327
c#	92,008	json	264
ruby	68,317	xml	258
c	43,487	yaml	180
rust	12,924	assembly	171
kotlin	11,284	powershell	162
sql	6,826	vba	157
pascal	6,299	lua	126
d	5,278	matlab	114
haskell	4,967	dart	107
scala	4,120	bash	105
html	2,777	http	99
shell	2,095	graphql	89
perl	2,009	svg	82
swift	1,907	vb.net	75
ocaml	1,894	groovy	63
csharp	1,891	Misc.	2,301

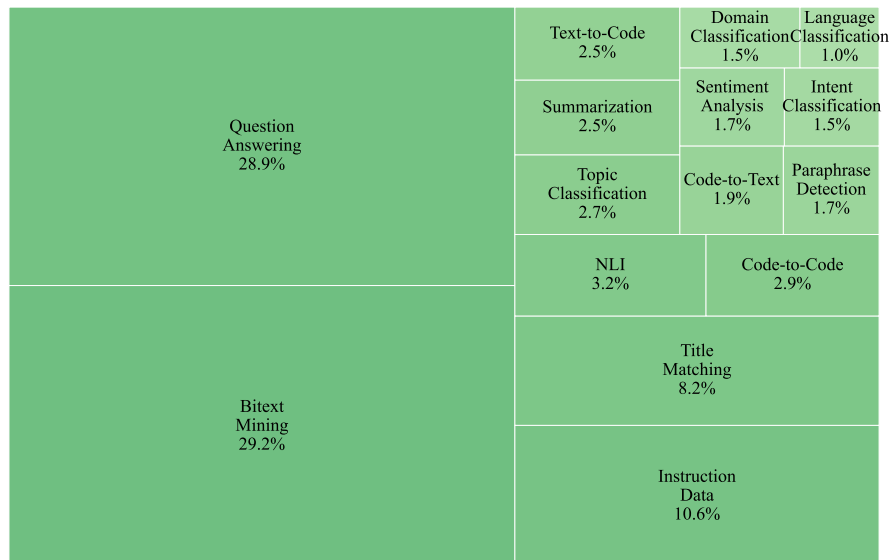


Figure 7. Task type distribution in our training data.

Table 8. Number of samples in our collected training dataset (part 1).

Name	Language	Format	Size	URL
Bitext Mining				
UNPC (Ziems et al., 2016)	6	Retrieval	2,922,245	huggingface.co/datasets/Helsinki-NLP/un_pc
ParaCrawl (Bañón et al., 2020)	30	Retrieval	10,684,184	paracrawl.eu/index.php
BactrianX Translation (Li et al., 2023a)	52	Clustering	491,282	huggingface.co/datasets/MB2UAI/Bactrian-X
Europarl (Koehn, 2005)	21	Clustering	477,566	huggingface.co/datasets/Helsinki-NLP/europarl
Question Answering				
WebFAQ (Dinzing et al., 2025)	49	Retrieval	4,368,504	huggingface.co/datasets/PaDaS-Lab/webfaq-retrieval
mMARCO (Bonifacio et al., 2021)	14	Retrieval	5,470,174	huggingface.co/datasets/unicamp-dl/mmarco
PAQ (Lewis et al., 2021)	en	Retrieval	938,771	huggingface.co/datasets/sentence-transformers/paq
SQuAD (Rajpurkar et al., 2016)	en	Retrieval	89,509	huggingface.co/datasets/rajpurkar/squad
Stack Exchange (Team, 2021b)	en	Retrieval	754,705	huggingface.co/datasets/flax-sentence-embeddings/stackexchange_titlebody_best_voted_answer_jsonl
Arguana (Wachsmuth et al., 2018)	en	Retrieval	22,848	huggingface.co/datasets/BeIR/arguana-generated-queries
Natural Questions (Kwiatkowski et al., 2019)	en	Retrieval	97,209	huggingface.co/datasets/sentence-transformers/natural-questions
HotpotQA (Yang et al., 2018)	en	Retrieval	120,528	huggingface.co/datasets/mteb/hotpotqa
ELI5 (Fan et al., 2019)	en	Retrieval	161,345	huggingface.co/datasets/Pavithree/eli5
FiQA2018 (Maia et al., 2018)	en	Retrieval	7,452	huggingface.co/datasets/mteb/figa
BioASQ (Tsatsaronis et al., 2015)	en	Retrieval	125,248	huggingface.co/datasets/BeIR/bioasq-generated-queries
NFCorpus (Boteva et al., 2016)	en	Retrieval	1,283	huggingface.co/datasets/mteb/nfcampus
TriviaQA (Joshi et al., 2017)	en	Retrieval	60,025	huggingface.co/datasets/sentence-transformers/trivia-qa-triplet
PubMedQA (Jin et al., 2019)	en	Retrieval	60,227	huggingface.co/datasets/qiaojin/PubMedQA
Amazon QA (Gupta et al., 2019)	en	Retrieval	59,340	github.com/amazonqa/amazonqa
MIRACL (Zhang et al., 2023b)	16	Retrieval	26,740	huggingface.co/datasets/miracrl/miracrl
Mr.TyDi (Zhang et al., 2021)	11	Retrieval	48,619	huggingface.co/datasets/mteb/mrtdy
MLDR (Chen et al., 2024)	13	Retrieval	40,264	huggingface.co/datasets/Shitao/MLDR
MKQA (Longpre et al., 2021)	26	Retrieval	69,287	huggingface.co/datasets/mteb/MKQARetrieval
StackOverflowQA (Li et al., 2025a)	en	Retrieval	13,820	huggingface.co/datasets/mteb/stackoverflow-qa
ProQA (Li et al., 2024b)	11	Retrieval	485,780	github.com/jordane95/procqa
Yahoo-Answers (Zhang et al., 2015)	en	Retrieval	196,645	huggingface.co/datasets/sentence-transformers/yahoo-answers
GooAQ (Khashabi et al., 2021)	en	Retrieval	473,876	github.com/allenai/gooaq
T2Ranking (Xie et al., 2023)	zh	Retrieval	85,521	huggingface.co/datasets/sentence-transformers/t2ranking
DuReader (He et al., 2018)	zh	Retrieval	78,023	huggingface.co/datasets/sentence-transformers/dureader
cMedQAv2 (Zhang et al., 2018)	zh	Retrieval	23,105	huggingface.co/datasets/sentence-transformers/cmedqa-v2
Huatuo.kgqa (Wang et al., 2025)	zh	Retrieval	53,835	huggingface.co/datasets/FreedomIntelligence/huatuo_knowledge_graph_qa
Huatuo_encqa (Wang et al., 2025)	zh	Retrieval	253,523	huggingface.co/datasets/FreedomIntelligence/huatuo_encyclopedia_qa
Multi CPR Medical (Long et al., 2022)	zh	Retrieval	62,085	github.com/Alibaba-NLP/Multi-CPR
HealthCareMagic (Li et al., 2023b)	en	Retrieval	78,626	github.com/KentOn-Li/ChatDoctor
MedicalQA_ru (Blinov, 2021)	ru	Retrieval	71,932	huggingface.co/datasets/blinoff/medical_qa_ru_data
Instruction Data				
Aya (Singh et al., 2024)	65	Retrieval	126,965	huggingface.co/datasets/CohereLabs/aya_dataset
MURI (Köksal et al., 2025)	194	Retrieval	720,782	huggingface.co/datasets/akoksal/muri-it
OASST2 (Köpf et al., 2023)	26	Retrieval	12,449	huggingface.co/datasets/OpenAssistant/oasst2
MultiAlpaca (Wei et al., 2023)	11	Retrieval	125,447	huggingface.co/datasets/DAMO-NLP-MT/multialpaca
WildChat (Zhao et al., 2024)	76	Retrieval	638,781	huggingface.co/datasets/allenai/WildChat-4.8M
M2Lingual (Maheshwary et al., 2025)	75	Retrieval	158,251	huggingface.co/datasets/ServiceNow-AI/M2Lingual
Natural Reasoning (Yuan et al., 2025)	en	Retrieval	845,682	huggingface.co/datasets/facebook/natural_reasoning
Infinity Instruct (Du et al., 2025)	en, zh	Retrieval	757,439	huggingface.co/datasets/BAAI/Infinity-Instruct
COIG (Bai et al., 2025)	zh	Retrieval	42,415	huggingface.co/datasets/m-a-p/COIG-CQIA
Medinstruct (Zhang et al., 2023c)	en	Retrieval	51,539	github.com/XZhang97666/AlpaCare
CodeFeedbackST (Li et al., 2025a)	137	Retrieval	115,971	huggingface.co/datasets/mteb/codefeedback-st
CodeFeedbackMT (Li et al., 2025a)	python	Retrieval	52,221	huggingface.co/datasets/mteb/codefeedback-mt
OpenOrca (Lian et al., 2023)	en	Retrieval	896,450	huggingface.co/datasets/Open-Orca/OpenOrca
MED12 (Muennighoff et al., 2025)	en	Retrieval	668,036	huggingface.co/datasets/GritLM/MED12
MedicalInstruction (Altaf, 2023)	en	Retrieval	75,268	huggingface.co/datasets/Mohammed-Altaf/medical-instruction-120k
Title Matching				
S2ORC-Title-Abstract (Lo et al., 2020)	en	Retrieval	250,000	huggingface.co/datasets/sentence-transformers/s2orc
CORD 19 (Wang et al., 2020)	en	Retrieval	373,674	huggingface.co/datasets/medalpaca/medical_meadow_cord19
Multi CPR ECom (Long et al., 2022)	zh	Retrieval	90,850	github.com/Alibaba-NLP/Multi-CPR
ESCI (Reddy et al., 2022)	en, ja, es	Retrieval	80,468	huggingface.co/datasets/tasksource/esci
CLIRMatrix (Sun & Duh, 2020)	137	Retrieval	3,275,561	github.com/ssun32/CLIRMatrix
NLI				
SNLI (Bowman et al., 2015)	en	Retrieval	54,585	huggingface.co/datasets/stanfordnlp/snli
MNLI (Williams et al., 2018)	en	Retrieval	112,075	huggingface.co/datasets/nyu-ml/multi_nli
ANLI (Nie et al., 2020)	en	Retrieval	18,801	huggingface.co/datasets/facebook/anli
XNLI (Conneau et al., 2018)	14	Retrieval	1,400,600	huggingface.co/datasets/mteb/xnli
OCNLI (Hu et al., 2020)	zh	Retrieval	6,616	huggingface.co/datasets/dirtycomputer/OCNLI
Code-to-Code				
xCodeEval Code2Code (Khan et al., 2024)	17	Retrieval	37,056	huggingface.co/datasets/NTU-NLP-sg/xCodeEval
xCodeEval Translation (Khan et al., 2024)	11	Clustering	500,000	huggingface.co/datasets/NTU-NLP-sg/xCodeEval
CodeSearchNet-ccr (Li et al., 2025a)	6	Retrieval	905,195	huggingface.co/datasets/CoIR-Retrieval/CodeSearchNet-ccr

Table 9. Number of samples in our collected training dataset (part 2).

Name	Language	Format	Size	URL
Topic Classification				
Arxiv Clustering P2P (Team, 2022a)	en	Clustering	83,476	huggingface.co/datasets/mteb/raw_arxiv
Arxiv Clustering S2S (Team, 2022a)	en	Clustering	83,486	huggingface.co/datasets/mteb/raw_arxiv
Biorxiv Clustering P2P (Team, 2022b)	en	Clustering	57,296	huggingface.co/datasets/mteb/raw_biorxiv
Biorxiv Clustering S2S (Team, 2022b)	en	Clustering	57,296	huggingface.co/datasets/mteb/raw_biorxiv
Medrxiv Clustering P2P (Team, 2022c)	en	Clustering	18,659	huggingface.co/datasets/mteb/raw_medrxiv
Medrxiv Clustering S2S (Team, 2022c)	en	Clustering	18,659	huggingface.co/datasets/mteb/raw_medrxiv
MLSUM Clustering (Scialom et al., 2020)	de, es, fr, ru	Clustering	325,739	huggingface.co/datasets/mteb/mlsum
TwentyNewsgroups (Lang, 1995)	en	Clustering	11,060	huggingface.co/datasets/SetFit/20_newsgroups
SIB200ClusteringS2S	205	Clustering	163,302	huggingface.co/datasets/mteb/sib200
Reddit Clustering P2P (Team, 2021d)	en	Clustering	80,000	huggingface.co/datasets/sentence-transformers/reddit-title-body
Reddit Clustering S2S (Geigle et al., 2021)	en	Clustering	58,141	github.com/UKPLab/TWEAC-qa-agent-selection/tree/master/data/reddit/train
Stack Exchange Clustering P2P (Team, 2021a)	en	Clustering	80,000	huggingface.co/datasets/flax-sentence-embeddings/stackexchange_title_body_jsonl
Stack Exchange Clustering S2S (Geigle et al., 2021)	en	Clustering	56,731	github.com/UKPLab/TWEAC-qa-agent-selection/tree/master/data/stackexchange/train
THUCNews (Sun et al., 2016)	zh	Clustering	100,000	huggingface.co/datasets/SirlyDreamer/THUCNews
TNews (Xu et al., 2020)	zh	Clustering	49,726	huggingface.co/datasets/C-MTEB/TNews-classification
CSL (Li et al., 2022)	zh	Clustering	100,000	huggingface.co/datasets/neuclir/csl
Summarization				
XSum (Narayan et al., 2018)	en	Retrieval	184,383	huggingface.co/datasets/EdinburghNLP/xsum
CNN_DM (Hermann et al., 2015)	en	Retrieval	100,000	huggingface.co/datasets/abisee/cnn_dailymail
MLSUM Retrieval (Scialom et al., 2020)	5	Retrieval	801,159	huggingface.co/datasets/mteb/mlsum
Sentence Compression (Filippova & Altun, 2013)	en	Retrieval	175,477	huggingface.co/datasets/sentence-transformers/sentence-compression
Text-to-Code				
OCGI (Majumdar et al., 2024)	python	Retrieval	1,052,849	huggingface.co/datasets/nvidia/OpenCodeGeneticInstruct
OpenCodeReasoning-2 (Ahmad et al., 2025)	python, cpp	Retrieval	16,632	huggingface.co/datasets/nvidia/OpenCodeReasoning-2
xCodeEval NL2Code (Khan et al., 2024)	17	Retrieval	51,072	huggingface.co/datasets/NTU-NLP-sg/xCodeEval
CosQA (Huang et al., 2021)	python	Retrieval	9,409	huggingface.co/datasets/mteb/cosqa
SyntheticText2SQL (Meyer et al., 2024)	sql	Retrieval	99,617	huggingface.co/datasets/mteb/synthetic-text2sql
Code-to-Text				
CodeSearchNet (Husain et al., 2019)	6	Retrieval	936,813	huggingface.co/datasets/CoIR-Retrieval/CodeSearchNet
Paraphrase Detection				
StackExchangeDupQuestions-S2S (Team, 2021c)	en	Retrieval	183,559	huggingface.co/datasets/sentence-transformers/stackexchange-duplicates
StackExchangeDupQuestions-P2P (Team, 2021c)	en	Retrieval	203,060	huggingface.co/datasets/sentence-transformers/stackexchange-duplicates
QQP (Sharma et al., 2019)	en	Retrieval	243,598	gluebenchmark.com/tasks
StackOverflowDupQuestions (Liu et al., 2018)	en	Retrieval	19,847	huggingface.co/datasets/mteb/stackoverflowdupquestions-reranking
PawsX (Yang et al., 2019)	7	Retrieval	216,219	huggingface.co/datasets/google-research-datasets/paws-x
Sentiment Analysis				
Amazon Polarity (McAuley & Leskovec, 2013)	en	Classification	100,000	huggingface.co/datasets/mteb/amazon_polarity
IMDb (Maas et al., 2011)	en	Classification	24,904	huggingface.co/datasets/mteb/imdb
Toxic Conversations (cjadams et al., 2019)	en	Classification	49,900	huggingface.co/datasets/mteb/toxic_conversations_50k
Amazon Counterfactual (O'Neill et al., 2021)	en, de, ja	Classification	14,870	huggingface.co/datasets/mteb/amazon_counterfactual
Amazon Reviews (McAuley & Leskovec, 2013)	6	Clustering	600,000	huggingface.co/datasets/mteb/amazon_reviews_multi
Emotion (Saravia et al., 2018)	en	Clustering	17,944	huggingface.co/datasets/mteb/emotion
Tweet Sentiment Extraction (Maggie, 2020)	en	Clustering	26,732	huggingface.co/datasets/mteb/tweet_sentiment_extraction
Intent Classification				
Massive Intent (FitzGerald et al., 2023)	51	Clustering	661,923	huggingface.co/datasets/mteb/amazon_massive_intent
MTOP Intent (Li et al., 2021)	6	Clustering	83,922	huggingface.co/datasets/mteb/mtop_intent
Banking77 (Casanueva et al., 2020)	en	Clustering	9,993	huggingface.co/datasets/mteb/banking77
Domain Classification				
Massive Scenario (FitzGerald et al., 2023)	51	Clustering	661,923	huggingface.co/datasets/mteb/amazon_massive_scenario
MTOP Domain (Li et al., 2021)	6	Clustering	83,922	huggingface.co/datasets/mteb/mtop_domain
Language Classification				
BactrianX Language Classification (Li et al., 2023a)	52	Clustering	491,405	huggingface.co/datasets/MBZUAI/Bactrian-X
Citation Prediction				
S2ORC-Title-Citation (Lo et al., 2020)	en	Retrieval	132,879	huggingface.co/datasets/sentence-transformers/s2orc
S2ORC-Abstract-Citation (Lo et al., 2020)	en	Retrieval	231,587	huggingface.co/datasets/sentence-transformers/s2orc
SPECTER (Cohan et al., 2020)	en	Retrieval	24,717	huggingface.co/datasets/sentence-transformers/specter
Linguistic Acceptability				
MELA (Zhang et al., 2024)	10	Classification	40,267	huggingface.co/datasets/Geralt-Targaryen/MELA
Scala (Nielsen, 2023)	9	Classification	128,471	huggingface.co/datasets/alexandrinst/scala
DaLa (Barnina et al., 2025)	da	Classification	6,508	huggingface.co/datasets/giannor/dala_large
Claim Verification				
FEVER (Thorne et al., 2018)	en	Retrieval	106,605	huggingface.co/datasets/mteb/fever
SciFact (Wadden et al., 2020)	en	Retrieval	859	huggingface.co/datasets/mteb/scifact
COLIEE (Kim et al., 2022)	en	Retrieval	454	www.modelscope.cn/datasets/sentence-transformers/coliee
STS				
STS12 (Agirre et al., 2012)	en	Retrieval	1,858	huggingface.co/datasets/mteb/sts12-sts
STS22 (Chen et al., 2022)	en	Retrieval	389	huggingface.co/datasets/mteb/sts22-crosslingual-sts
STSBenchmark (May, 2021)	en	Retrieval	3,297	huggingface.co/datasets/mteb/stsbenchmark-sts
STS22-Crosslingual (Chen et al., 2022)	7	Retrieval	1,469	huggingface.co/datasets/mteb/sts22-crosslingual-sts
BQ (Xiao et al., 2023)	zh	Retrieval	2,436	huggingface.co/datasets/C-MTEB/BQ

B. Details on MTEB Evaluation

The Massive Text Embedding Benchmark (MTEB) is widely recognized as the de facto standard for the comprehensive evaluation of text embedding models. Originally introduced by [Muennighoff et al. \(2023\)](#), it was vastly expanded into the Massive Multilingual Text Embedding Benchmark (MMTEB) through a large-scale, open-science collaboration ([Enevoldsen et al., 2025](#)). This community-driven effort has established a rigorous and diverse evaluation framework, encompassing over 500 quality-controlled tasks that span more than 250 languages and a wide array of domains.

The significance of MTEB lies in its unprecedented scale and diversity, which addresses the critical limitations of previous benchmarks that were often constrained to a few languages (mostly English), specific domains (e.g., news), or a single task type (e.g., retrieval). To provide a holistic assessment of a model’s capabilities, MTEB organizes its evaluation tasks into ten distinct categories:

- **Retrieval:** Assesses a model’s ability to find relevant documents from a large corpus for a given query.
- **Reranking:** Measures the ability to reorder a given list of candidate documents by their relevance to a query.
- **Classification:** Evaluates performance on standard text classification tasks (e.g., sentiment analysis, topic classification).
- **Clustering:** Tests how well embeddings group semantically similar documents together.
- **Pair Classification:** Involves predicting the relationship between a pair of texts (e.g., paraphrase detection, natural language inference).
- **Semantic Textual Similarity (STS):** Measures the ability to predict the degree of semantic similarity between two sentences on a continuous scale.
- **Bitext Mining:** Assesses the ability to identify translated sentence pairs from a collection of sentences in two languages.
- **Summarization:** Evaluates the semantic similarity between a model-generated summary and a reference summary.
- **Instruction Reranking:** A more challenging reranking variant where the model must follow a detailed natural language instruction to determine relevance.
- **Multilabel Classification:** A classification variant where each document can be assigned multiple labels.

The hundreds of tasks are further organized into benchmarks, which are curated subsets of tasks grouped by language, domain, or a combination of both. This includes language-specific benchmarks such as English, Chinese, and Russian; domain-specific benchmarks such as Code and Medical; and aggregated benchmarks like Multilingual, European, and Scandinavian, which test performance across a broad and diverse set of languages. This hierarchical structure allows for both a fine-grained analysis of a model’s performance on a specific language or domain and a high-level view of its overall multilingual and multi-domain capabilities.

In this work, we leverage the breadth of MTEB to provide a robust and thorough evaluation of our models. We evaluate on **17 benchmarks**, totaling **430 unique tasks**: Multilingual, Code, Medical, English, Russian, French, German, Polish, Dutch, Indic, Persian, Chinese, Japanese, Korean, Vietnamese, European, and Scandinavian. This extensive evaluation allows for a robust and fine-grained assessment of our models’ capabilities, directly supporting our claims of multilingual inclusivity and broad domain competence. The complete list of tasks used in our evaluation is detailed in Tables 10-14.

Table 10. MTEB tasks evaluated in this work: Multilingual, Code, and Medical benchmarks.

Category	Tasks
Benchmark: Multilingual	
Bitext Mining	BornholmBitextMining, BibleNLPBitextMining, BUCC.v2, DiaBlaBitextMining, FloresBitextMining, IN22GenBitextMining, IndicGenBenchFloresBitextMining, NollySentiBitextMining, NorwegianCourtsBitextMining, NTREXBitextMining, NusaTranslationBitextMining, NusaXBitextMining, Tatoeba
Classification	AfriSentiClassification, AmazonCounterfactualClassification, BulgarianStoreReviewSentimentClassification, CSFD-SKMReviewSentimentClassification, CataloniaTweetClassification, CyrillicTurkicLangClassification, CzechProductReviewSentimentClassification, DBpediaClassification, DalajClassification, EstonianValenceClassification, FilipinoShopeeReviewsClassification, FinancialPhrasebankClassification, GreekLegalCodeClassification, GujaratiNewsClassification, IndicLangClassification, IndonesianIdClickbaitClassification, IsiZuluNewsClassification, ItaCaseholdClassification, KorSarcasmClassification, KurdishSentimentClassification, MacedonianTweetSentimentClassification, MasakhaNEWSClassification, MassiveIntentClassification, MultiHateClassification, NepaliNewsClassification, NordicLangClassification, NusaParagraphEmotionClassification, NusaX-senti, OdiaNewsClassification, PAC, PoemSentimentClassification, PolEmo2.0-OUT, PunjabiNewsClassification, ScalaClassification, SentimentAnalysisHindi, SinhalaNewsClassification, SiswatiNewsClassification, SlovakMovieReviewSentimentClassification, SwahiliNewsClassification, SwissJudgementClassification, ToxicConversationsClassification, TswanaNewsClassification, TweetTopicSingleClassification
Clustering	AlloProfClusteringS2S.v2, ArXivHierarchicalClusteringP2P, ArXivHierarchicalClusteringS2S, BigPatentClustering.v2, BiorxivClusteringP2P.v2, CLSCLusteringP2P.v2, HALClusteringS2S.v2, MasakhaNEWSClusteringS2S, MedrxivClusteringP2P.v2, PlscClusteringP2P.v2, RomaniBibleClustering, SIB200ClusteringS2S, StackExchangeClustering.v2, SwednClusteringP2P, WikiCitiesClustering, WikiClusteringP2P.v2
Instruction Reranking	Core17InstructionRetrieval, News21InstructionRetrieval, Robust04InstructionRetrieval
Multilabel Classification	BrazilianToxicTweetsClassification, CEDRClassification, KorHateSpeechMLClassification, MalteseNewsClassification, MultiEURLEXMultilabelClassification
Pair Classification	ArmenianParaphrasePC, CTKFactsNLI, OpusparcusPC, PawsXPairClassification, PpcPC, RTE3, SprintDuplicateQuestions, TERRa, TwitterURLCorpus, XNLI, indonli
Reranking	AlloprofReranking, RuBQReranking, T2Reranking, VoyageMMarcoReranking, WebLINXCandidatesReranking, WikipediaRerankingMultilingual
Retrieval	AILAStatutes, ArguAna, BelebeleRetrieval, CovidRetrieval, HagridRetrieval, LEMBPaskeyRetrieval, LegalBenchCorporateLobbying, MIRACLRetrievalHardNegatives, MLQARetrieval, SCIDOCS, SpartQA, StackOverflowQA, StatcanDialogueDatasetRetrieval, TRECCOVID, TempReasonL1, TwitterHjerneRetrieval, WikipediaRetrievalMultilingual, WinoGrande
STS	FaroeeseSTS, FinParaSTS, GermanSTSBenchmark, IndicCrosslingualSTS, JSICK, SICK-R, STS12, STS13, STS14, STS15, STS17, STS22.v2, STSB, STSBenchmark, STSES, SemRel24STS
Benchmark: Code	
Retrieval	AppsRetrieval, CodeEditSearchRetrieval, CodeFeedbackMT, CodeFeedbackST, CodeSearchNetCCRetrieval, CodeSearchNetRetrieval, CodeTransOceanContest, CodeTransOceanDL, CosQA, COIRCodeSearchNetRetrieval, StackOverflowQA, SyntheticText2SQL
Benchmark: Medical	
Clustering	MedrxivClusteringP2P.v2, MedrxivClusteringS2S.v2
Retrieval	CUREv1, NFCorpus, TRECCOVID, TRECCOVID-PL, SciFact, SciFact-PL, MedicalQARetrieval, PublicHealthQA, CmedqaRetrieval
Reranking	CMedQAv2-reranking

Table 11. MTEB tasks evaluated in this work: Russian, French, German, and Polish benchmarks.

Category	Tasks
Benchmark: Russian	
Classification	GeoreviewClassification, HeadlineClassification, InappropriatenessClassification, KinopoiskClassification, MassiveIntentClassification, MassiveScenarioClassification, RuReviewsClassification, RuSciBenchGRNTIClassification, RuSciBenchOECDClassification
Clustering	GeoreviewClusteringP2P, RuSciBenchGRNTIClusteringP2P, RuSciBenchOECDClusteringP2P
Multiclass Classification	CEDRClassification, SensitiveTopicsClassification
Pair Classification	TERRa
Reranking	MIRACLReranking, RuBQReranking
Retrieval	MIRACLRetrievalHardNegatives.v2, RiaNewsRetrievalHardNegatives.v2, RuBQRetrieval
STS	RUParaPhraserSTS, STS22, RuSTSBenchmarkSTS
Benchmark: French	
Classification	AmazonReviewsClassification, MasakhaNEWSClassification, MassiveIntentClassification, MassiveScenarioClassification, MTOPDomainClassification, MTOPIntentClassification
Clustering	AlloProfClusteringP2P, AlloProfClusteringS2S, HALClusteringS2S, MasakhaNEWSClusteringP2P, MasakhaNEWSClusteringS2S, MLSUMClusteringP2P, MLSUMClusteringS2S
Pair Classification	PawsXPairClassification
Reranking	AlloprofReranking, SyntecReranking
Retrieval	AlloprofRetrieval, BSARDRetrieval, MintakaRetrieval, SyntecRetrieval, XPQARetrieval
STS	SICKFr, STSBenchmarkMultilingualSTS, STS22
Summarization	SummEvalFr
Benchmark: German	
Classification	AmazonCounterfactualClassification, AmazonReviewsClassification, MTOPDomainClassification, MTOPIntentClassification, MassiveIntentClassification, MassiveScenarioClassification
Clustering	BlurbsClusteringP2P, BlurbsClusteringS2S, TenKGnadClusteringP2P, TenKGnadClusteringS2S
Pair Classification	FalseFriendsGermanEnglish, PawsXPairClassification
Reranking	MIRACLReranking
Retrieval	GermanQuAD-Retrieval, GermanDPR, XMarket, GerDaLIR
STS	GermanSTSBenchmark, STS22
Benchmark: Polish	
Classification	AllegroReviews, CBD, MassiveIntentClassification, MassiveScenarioClassification, PolEmo2.0-IN, PolEmo2.0-OUT, PAC
Clustering	EightTagsClustering, PlscClusteringS2S, PlscClusteringP2P
Pair Classification	CDSC-E, PpcPC, PSC, SICK-E-PL
STS	CDSC-R, SICK-R-PL, STS22

Table 12. MTEB tasks evaluated in this work: Dutch, Indic, and Persian benchmarks.

Category	Tasks
Benchmark: Dutch	
Classification	DutchBookReviewSentimentClassification.v2, MassiveIntentClassification, MassiveScenarioClassification, SIB200Classification, MultiHateClassification, VaccinChatNLClassification, DutchColaClassification, DutchGovernmentBiasClassification, DutchSarcasticHeadlinesClassification, DutchNewsArticlesClassification, OpenTenderClassification, IconclassClassification
Pair Classification	SICKNLPairClassification, XLWICNLPairClassification
Multiclass Classification	CovidDisinformationNLMultiLabelClassification, MultiEURLEXMultilabelClassification, VABBMultilabelClassification
Clustering	DutchNewsArticlesClusteringS2S, DutchNewsArticlesClusteringP2P, SIB200ClusteringS2S, VABBClusteringS2S, VABBClusteringP2P, OpenTenderClusteringS2S, OpenTenderClusteringP2P, IconclassClusteringS2S
Reranking	WikipediaRerankingMultilingual
Retrieval	ArguAna-NL.v2, SCIDOCS-NL.v2, SciFact-NL.v2, NFCorpus-NL.v2, BelebeleRetrieval, WebFAQRetrieval, DutchNewsArticlesRetrieval, bBSARDNLRetrieval, LegalQANLRetrieval, OpenTenderRetrieval, VABBRetrieval, WikipediaRetrievalMultilingual
STS	SICK-NL-STS, STSBenchmarkMultilingualSTS
Benchmark: Indic	
Bitext Mining	IN22ConvBitextMining, IN22GenBitextMining
Clustering	SIB200ClusteringS2S
Classification	BengaliSentimentAnalysis, GujaratiNewsClassification, HindiDiscourseClassification, SentimentAnalysisHindi, MalayalamNewsClassification, MTOPIntentClassification, MultiHateClassification, TweetSentimentClassification, NepaliNewsClassification, PunjabiNewsClassification, SanskritShlokasClassification, UrduRomanSentimentClassification
Pair Classification	XNLI
Retrieval	BelebeleRetrieval, XQuADRetrieval
Reranking	WikipediaRerankingMultilingual
STS	IndicCrosslingualSTS
Benchmark: Persian	
Classification	PersianFoodSentimentClassification, SynPerChatbotConvSAClassification, SynPerChatbotConvSAToneChatbotClassification, SynPerChatbotConvSAToneUserClassification, SynPerChatbotSatisfactionLevelClassification, SynPerTextToneClassification.v3, SIDClassification.v2, DeepSentiPers.v2, PersianTextEmotion.v2, NLPTwitterAnalysisClassification.v2, DigikalamagClassification, MassiveIntentClassification, MassiveScenarioClassification, StyleClassification, PerShopDomainClassification, PerShopIntentClassification
Clustering	BeytooteClustering, DigikalamagClustering, HamshahriClustering, NLPTwitterAnalysisClustering, SIDClustering
Pair Classification	FarsTail, SynPerChatbotRAGFAQPC, FarsiParaphraseDetection, SynPerTextKeywordsPC, SynPerQAPC, ParsinluEntail, ParsinluQueryParaphPC
Reranking	MIRACLReRanking, WikipediaRerankingMultilingual
Retrieval	SynPerQARetrieval, SynPerChatbotRAGFAQRetrieval, PersianWebDocumentRetrieval, WikipediaRetrievalMultilingual, MIRACLRetrievalHardNegatives, HotpotQA-FaHardNegatives, MSMARCO-FaHardNegatives, NQ-FaHardNegatives, ArguAna-Fa.v2, FiQA2018-Fa.v2, QuoraRetrieval-Fa.v2, SCIDOCS-Fa.v2, SciFact-Fa.v2, TRECCOVID-Fa.v2, FEVER-FaHardNegatives, NeuCLIR2023RetrievalHardNegatives, WebFAQRetrieval
STS	Farsick, SynPerSTS
Bitext Mining	SAMSumFa, SynPerChatbotSumSRetrieval, SynPerChatbotRAGSumSRetrieval

Table 13. MTEB tasks evaluated in this work: English, Scandinavian, and European benchmarks.

Category	Tasks
Benchmark: English	
Classification	AmazonCounterfactualClassification, Banking77Classification, ImdbClassification, MTPDomainClassification, MassiveIntentClassification, MassiveScenarioClassification, ToxicConversationsClassification, TweetSentimentExtractionClassification
Clustering	ArXivHierarchicalClusteringP2P, ArXivHierarchicalClusteringS2S, BiorxivClusteringP2P.v2, MedrxivClusteringP2P.v2, MedrxivClusteringS2S.v2, StackExchangeClustering.v2, StackExchangeClusteringP2P.v2, TwentyNewsgroupsClustering.v2
Pair Classification	SprintDuplicateQuestions, TwitterSemEval2015, TwitterURLCorpus
Reranking	AskUbuntuDupQuestions, MindSmallReranking
Retrieval	ArguAna, CQADupstackGamingRetrieval, CQADupstackUnixRetrieval, ClimateFEVERHardNegatives, FEVERHardNegatives, FiQA2018, HotpotQAHardNegatives, SCIDOCS, TRECCOVID, Touche2020Retrieval.v3
STS	BIOSSES, SICK-R, STS12, STS13, STS14, STS15, STSBenchmark, STS17, STS22.v2
Summarization	SummEvalSummarization.v2
Benchmark: Scandinavian	
Bitext Mining	BornholmBitextMining, NorwegianCourtsBitextMining
Classification	AngryTweetsClassification, DanishPoliticalCommentsClassification, DalajClassification, DKHateClassification, LccSentimentClassification, MassiveIntentClassification, MassiveScenarioClassification, NordicLangClassification, NoRecClassification, NorwegianParliamentClassification, ScalaClassification, SwedishSentimentClassification, SweRecClassification
Retrieval	DanFeverRetrieval, NorQuadRetrieval, SNLRetrieval, SwednRetrieval, SweFaqRetrieval, TV2Nordretrieval, TwitterHjerneRetrieval
Clustering	SNLHierarchicalClusteringS2S, SNLHierarchicalClusteringP2P, SwednClusteringP2P, SwednClusteringS2S, VGHierarchicalClusteringS2S, VGHierarchicalClusteringP2P
Benchmark: European	
Bitext Mining	BornholmBitextMining, BibleNLPBitextMining, BUCC.v2, DiaBlaBitextMining, FloresBitextMining, NorwegianCourtsBitextMining, NTREXBitextMining
Classification	BulgarianStoreReviewSentimentClassification, CzechProductReviewSentimentClassification, GreekLegalCodeClassification, DBpediaClassification, FinancialPhrasebankClassification, PoemSentimentClassification, ToxicChatClassification, ToxicConversationsClassification, EstonianValenceClassification, ItaCaseholdClassification, AmazonCounterfactualClassification, MassiveScenarioClassification, MultiHateClassification, ScalaClassification, SwissJudgementClassification, TweetSentimentClassification, CBD, PolEmo2.0-OUT, CSFDSKMovieReviewSentimentClassification, DalajClassification
Clustering	WikiCitiesClustering, RomaniBibleClustering, BigPatentClustering.v2, BiorxivClusteringP2P.v2, AlloProfClusteringS2S.v2, HALClusteringS2S.v2, SIB200ClusteringS2S, WikiClusteringP2P.v2
Retrieval	StackOverflowQA, TwitterHjerneRetrieval, LegalQuAD, ArguAna, HagridRetrieval, LegalBenchCorporateLobbying, LEMBPasskeyRetrieval, SCIDOCS, SpartQA, TempReasonL1, WinoGrande, AlloprofRetrieval, BelebeleRetrieval, StatcanDialogueDatasetRetrieval, WikipediaRetrievalMultilingual
Instruction Reranking	Core17InstructionRetrieval, News21InstructionRetrieval, Robust04InstructionRetrieval
Multiclass Classification	MalteseNewsClassification, MultiEURLEXMultilabelClassification
Pair Classification	CTKfactsNLI, SprintDuplicateQuestions, OpusparcusPC, RTE3, XNLI, PSC
Reranking	WebLINXCandidatesReranking, AlloprofReranking, WikipediaRerankingMultilingual
STS	SICK-R, STS12, STS14, STS15, STSBenchmark, FinParaSTS, STS17, SICK-R-PL, STSES

Table 14. MTEB tasks evaluated in this work: Chinese, Japanese, Korean, and Vietnamese benchmarks.

Category	Tasks
Benchmark: Chinese	
Retrieval	T2Retrieval, MMarcoRetrieval, DuRetrieval, CovidRetrieval, CmedqaRetrieval, EcomRetrieval, MedicalRetrieval, VideoRetrieval
Reranking	T2Reranking, MMarcoReranking, CMedQAv1-reranking, CMedQAv2-reranking
Pair Classification	Ocnli, Cmnli
Clustering	CLSClusteringS2S, CLSClusteringP2P, ThuNewsClusteringS2S, ThuNewsClusteringP2P
Classification	TNews, IFlyTek, Waimai, OnlineShopping, JDReview, MultilingualSentiment, MultilingualSentiment
STS	LCQMC, PAWSX, AFQMC, QBQTC, ATEC, BQ, STSB
Benchmark: Japanese	
Clustering	LivedoorNewsClustering.v2, MewsC16JaClustering, SIB200ClusteringS2S
Classification	AmazonReviewsClassification, AmazonCounterfactualClassification, MassiveIntentClassification, MassiveScenarioClassification, JapaneseSentimentClassification, SIB200Classification, WRIMEClassification
Retrieval	JaquetRetrieval, MrTidyRetrieval, JaGovFaqRetrieval, NLPJournalTitleAbsRetrieval.V2, NLPJournalTitleIntroRetrieval.V2, NLPJournalAbsIntroRetrieval.V2, NLPJournalAbsArticleRetrieval.V2, JaCWIRRetrieval, MIRACLRetrieval, MintakaRetrieval, MultiLongDocRetrieval
Reranking	ESCIReranking, JQAReranking, JaCWIRReranking, MIRACLReranking, MultiLongDocReranking
STS	JSTS, JSICK
Benchmark: Korean	
Classification	KLUE-TC
Reranking	MIRACLReranking
Retrieval	MIRACLRetrieval, Ko-StrategyQA
STS	KLUE-STs, KorSTS
Benchmark: Vietnamese	
Retrieval	ArguAna-VN, SciFact-VN, ClimateFEVER-VN, FEVER-VN, DBPedia-VN, NQ-VN, HotpotQA-VN, MSMARCO-VN, TRECCOVID-VN, FiQA2018-VN, NFCorpus-VN, SCIDOCS-VN, Touche2020-VN, Quora-VN, CQADupstackAndroid-VN, CQADupstackGis-VN, CQADupstackMathematica-VN, CQADupstackPhysics-VN, CQADupstackProgrammers-VN, CQADupstackStats-VN, CQADupstackTex-VN, CQADupstackUnix-VN, CQADupstackWebmasters-VN, CQADupstackWordpress-VN
Classification	Banking77VNCClassification, EmotionVNCClassification, AmazonCounterfactualVNCClassification, MTOPDomainVNCClassification, TweetSentimentExtractionVNCClassification, ToxicConversationsVNCClassification, ImdbVNCClassification, MTOPIntentVNCClassification, MassiveScenarioVNCClassification, MassiveIntentVNCClassification, AmazonReviewsVNCClassification, AmazonPolarityVNCClassification
Pair Classification	SprintDuplicateQuestions-VN, TwitterSemEval2015-VN, TwitterURLCorpus-VN
Clustering	TwentyNewsgroupsClustering-VN, RedditClusteringP2P-VN, StackExchangeClusteringP2P-VN, StackExchangeClustering-VN, RedditClustering-VN
Reranking	SciDocsRR-VN, AskUbuntuDupQuestions-VN, StackOverflowDupQuestions-VN
STS	BIOSSES-VN, SICK-R-VN, STSBenchmark-VN

C. Training Details

C.1. Training Hyperparameters

We train the models with the contrastive loss

$$\mathcal{L} = -\log \frac{e^{s(q_i, d_i^+)/\tau}}{e^{s(q_i, d_i^+)/\tau} + \sum_{j=1}^n e^{s(q_i, d_{i,j}^-)/\tau}}, \quad (5)$$

where q is the query, d^+ , d^- are positive and negative documents, and $s(\cdot)$ is cosine similarity. The temperature τ is set to 0.5, and the number of hard negatives n is set to 7 (except for classification data, where $n = 1$). The loss coefficients $c_{l,d'}$ in Equation (3) are set to $c_{*,d'} = \frac{1}{\sqrt{d_{\text{model}}/d'}}$ for all layers. The models are trained with AdamW (Loshchilov & Hutter, 2019), ZeRO stage 2 (Rajbhandari et al., 2020), and Flash Attention 2 enabled (Dao, 2024). We set the input sequence length to 1024 during training, and the other hyperparameters are given in Table 15.

Table 15. Training hyperparameters.

Size	Learning Rate	Data Parallel	Batch Size
0.6B	1e-5	32	16
1.7B	9e-6	32	16
4B	8e-6	64	8
8B	7e-6	128	4

Near the end of training, we save checkpoints at intervals of 500 steps and merge the weights of the last 5 checkpoints. Ablation experiments on a subset of four benchmarks show that this slightly but consistently improves model performance (Table 16).

Table 16. Ablation results on the effectiveness of checkpoint merging.

Model	Multilingual ⁽¹¹⁸⁾	English ⁽³⁷⁾	Code ⁽¹⁰⁾	Medical ⁽¹¹⁾
W.o. merging	63.02	70.48	74.71	59.18
W. merging	63.03	70.52	74.75	59.33

C.2. Two-stage Training

In the two-stage training setting, we use MMARCO, WebFAQ, CLIRMatrix, ParaCrawl, OCGI, CodeSearchNet, and CodeSearchNet-CCR for first-stage training, totaling 26.7 million samples. In the second stage, we sample at most 100 thousand queries from each data source (different language subsets within a dataset are considered to be different sources), and train the models on 8.3 million samples. As demonstrated in Table 17, this amount of training data is an order of magnitude smaller than the data used to train other state-of-the-art multilingual embedding models. Paired with our fully-open training data, this recipe represents a significant step forward in promoting open and reproducible embedding model research.

Table 17. Comparison of training data size (in million) among multilingual embedding models. *EmbeddingGemma’s training data size is estimated based on the token count reported by Vera et al. (2025) (314B, 20B) and a context length of 2048. This model additionally underwent 2T tokens of encoder-decoder training from the causal model before the two-stage finetuning shown in the table.

Model	Open-data	First Stage	Second Stage
Qwen3-Embedding	×	150	12
EmbeddingGemma*	×	153	10
KaLM-Embedding	✓	100	5
Ours	✓	27	8

In Table 18, we present the ablation results on the impact of two-stage training conducted with the 0.6B model. We find that while the additional retrieval pre-finetuning improves performance on multilingual and English natural language

performance, it comes at the cost of reduced performance on code and medical benchmarks. However, this may be related to the composition of our pre-finetuning data, and deserves in-depth investigation in the future.

Table 18. Ablation results on the effectiveness of two-stage training with the 0.6B model.

Model	Multilingual ⁽¹¹⁸⁾	English ⁽³⁷⁾	Code ⁽¹⁰⁾	Medical ⁽¹¹⁾
Two-stage	63.02	70.48	74.71	59.18
Second-stage-only	62.34	70.29	75.01	59.56

D. Additional Results

D.1. 3D-ML Training on Individual Languages

In complement to the experiments in Section 5, we train two additional sets of models on Vietnamese (1M training samples) and Persian (350K training samples), respectively, starting from the same stage-1 0.6B checkpoint used in the main experiments. The results in Table 19 show that 3D-ML consistently outperforms the baseline when training on low-resource languages alone, confirming its effectiveness and wide applicability.

Table 19. Comparison of baseline (140M) and 3D-ML (pruned to 140M after training) models trained individually on Vietnamese and Persian.

Model	Vietnamese	Persian
Baseline	40.97	53.88
3D-ML	41.78	54.31

D.2. MEL in Language Modeling

To explore the full potential of MEL, we apply it to training causal language models. Specifically, we select Qwen2.5-0.5B-Base (Yang et al., 2024) as the backbone model to avoid benchmark contamination and saturation, and train two models on the allenai/tulu-3-sft-olmo-2-mixture³ data: 1) baseline SFT, and 2) training with MEL. For the second mode, we factorize the embedding matrix during evaluation, reducing its parameter count to 0.4B.

For evaluation, we adopt three widely used benchmarks: GSM8K (Cobbe et al., 2021), MMLU-Pro (Wang et al., 2024b), and WinoGrande (Sakaguchi et al., 2020). Interestingly, the results in Table 20 indicate that performance increases despite the smaller parameter count. We hypothesize that for smaller LMs with a disproportionately large embedding layer, MEL acts as an effective regularizer and improves generalization.

Table 20. Comparison of training Qwen2.5-0.5B-Base with and without MEL. All results are evaluated in 0-shot.

Model	GSM8K	MMLU-Pro	Winogrande
Baseline (0.5B)	40.8	9.9	47.1
MEL (0.4B)	43.5	11.1	50.6

D.3. Efficiency Analysis

In Table 21, we report the peak GPU memory consumption and token throughput of layer-pruned models in Figure 4, measured on a single A100 GPU. These results quantify the substantial, practical efficiency gains from 3D-ML. For instance, pruning the model from 28 layers to just a single layer increases throughput by over 13x (from 43k to 583k tokens/s), while reducing active parameters by over 70% and peak memory by 24%. When viewed alongside the performance curves in Figure 4, this data creates a concrete Pareto frontier, directly connecting model performance to tangible deployment metrics like latency (inversely related to throughput) and memory. However, we note that the fixed CUDA context overhead may present a confounding factor on memory measurements.

³<https://huggingface.co/datasets/allenai/tulu-3-sft-olmo-2-mixture>

Table 21. peak GPU memory consumption and token throughput of layer-pruned models in Figure 4.

# Layer	# Parameters (M)	Peak Memory (GB)	Throughput (token/s)
1	172	3.35	583831
2	187	3.63	318393
4	219	3.68	224050
8	282	3.80	148569
16	407	4.04	82832
24	533	4.27	58991
28	596	4.39	43218

E. Limitations

While our work demonstrates strong empirical performance and practical efficiency gains, several limitations remain:

Potential entanglement of method and data contributions Our results combine improvements from both the proposed 3D-ML framework and a newly curated multilingual dataset. Although we provide ablations to disentangle these factors on the smaller models, isolating their individual contributions remains challenging at scale. We hope our large-scale data could provide a platform to standardize comparisons of future training methods in this regard.

Potential dependence on base model architecture and complexity of hyperparameter choices Our models are built on Qwen3 causal architectures. While the 3D-ML framework is conceptually general and validated in a small-scale experiment on EuroBERT, generalization to more model architectures remains an open question.

The integration of MEL, MLL, and MRL introduces additional design choices (e.g., layer selection, rank schedules, and dimension sets). Although we provide reasonable defaults, the framework may require careful tuning for optimal performance in new settings.

Benchmark coverage and real-world deployment Despite covering 430 tasks and 17 benchmarks, our claims of inclusivity is bound by the coverage of benchmarks available in MTEB, which remains limited relative to the full long-tail of human languages. Moreover, the scores on MTEB benchmarks may not accurately reflect downstream application performance such as retrieval systems or RAG pipelines, and we suggest further validation before deploying our models in real-world applications.

Compute requirements for large models While 3D-ML improves efficiency at training and inference time, training and deploying the largest models (e.g., 8B) still requires substantial computational resources, which may limit reproducibility and accessibility.